KCC CODE CLUB · GAME MAKERS

Week 7

Operators, My Blocks and Levels

Week 7 of 8

Name _____

Today's big idea

Two ideas today.

Operators let you do maths and ask complicated questions. `(score) + 10` · `(score) > 10` · `<this> and <that>` · `join [Level ] (level)`

My Blocks let you take a group of blocks you keep rewriting, give it a name, and use it as one single block from then on.

> *"How do you make a cup of tea?"* — seven steps. Draw a box around them and call it **MAKE TEA**. Now "make tea for four people" is `repeat 4 [MAKE TEA]` instead of 28 steps.

That's called **abstraction**, and it's one of the most powerful ideas in all of programming.

Words to know

Word	What it means
operator	A block that does maths or asks a question
My Block	A block you invent yourself, made of other blocks
define	The hat block where you say what your new block does
abstraction	Naming a group of steps so you can forget the details

> **Dragging tip:** building `<<(score) > 10> and <(lives) > 1>>` inside a tiny slot is horrible. Build the whole thing in an **empty space** first, then drag the finished assembly into the `if`.

Class activity — Quiz Sprite

```
ask [What is 7 x 6?] and wait                    SENSING
if <(answer) = 42> then                          CONTROL
    say [Correct!] for 2 seconds                 LOOKS
    change score by 1                            VARIABLES
```

```
else                                                    CONTROL
    say (join [No, it was ] [42]) for 2 seconds         LOOKS
```

Add two more questions. Notice how much you're copying and pasting. Bonus challenge 2 fixes that.

Your project — it gets harder as you get better

- [ ] A `speed` variable. Replace the fixed `wait 0.02` with `wait (speed) seconds`
- [ ] Every 10 points: level up and multiply `speed` by 0.85

```
forever                                                 CONTROL
    if <(score) > ((level) * 10)> then
        change level by 1                               VARIABLES
        set speed to ((speed) * 0.85)
        say (join [Level ] (level)) for 1 seconds       LOOKS
    wait 0.1 seconds                                    CONTROL
```

- [ ] A **rare bonus item** worth 5 points that only shows up sometimes:

```
if <(pick random 1 to 5) = 1> then                      CONTROL
    show                                                LOOKS
else                                                    CONTROL
    hide                                                LOOKS
```

- [ ] *Homework:* make a `reset object` My Block and use it on every falling sprite
- [ ] **Save it**

Homework

1. A rare treasure. If you didn't finish the bonus item, do it now.

2. Tidy up. Find code you've written more than once and turn it into a My Block.

3. Think about the showcase. At the showcase, your family comes to see your game. Have two sentences ready: *what your game is*, and *the hardest bug you fixed*.

Bonus challenges

1. A power-up that makes your player wider for five seconds, then back to normal.
2. **Make your own block.** In your quiz, make a My Block called `celebrate` — say Correct!, grow, play a cheer, shrink back. Swap it in for all three copies. **Change the definition once → all three change.**
3. Use `and` or `or` for a combo condition — a bonus that only counts on a high level.
4. **Really hard:** replace your duplicated falling sprites with **clones**. Look up `create clone of myself`, `when I start as a clone`, `delete this clone`. Nobody is expected to get this one.

If something breaks

- **Game freezes on level up?** `speed` hit zero or went negative. Put a floor on it.
- **Level goes up hundreds of times?** The `if` keeps firing. Make sure `level` is in the condition.
- **Can't drop an operator in?** Build it in empty space first, then drag it in.
- **My Block missing on another sprite?** My Blocks belong to one sprite only.

Stuck? Put your hand up — that is what we are here for. And always save before you close the lid.